TakeMeter — Fine-Tuning Starter Notebook

AI201 · Project 3

This notebook walks you through fine-tuning a text classifier on your annotated dataset and comparing it to a zero-shot baseline.

What this notebook does for you (infrastructure):

- Tokenizes your dataset and prepares it for training
- Runs the fine-tuning pipeline with DistilBERT
- Computes evaluation metrics and generates a confusion matrix
- Runs the Groq baseline and compares both models

What you do (the actual work):

- Collect and annotate your 200+ examples (done before opening this notebook)
- Define your label map and upload your CSV
- Write your Groq classification prompt using your label definitions
- Analyze the output and write your evaluation report

Before you start: Make sure you are using a T4 GPU runtime.
Go to **Runtime → Change runtime type → T4 GPU**, then click Save.

```
# Install any dependencies not pre-installed on Colab
!pip install -q groq python-dotenv
print("✅ Dependencies ready")
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143.7/143.7 kB 8.8 MB/s eta 0:00:00
✅ Dependencies ready
```

```
import pandas as pd
import numpy as np
import json
import time

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, classification_report,
    confusion_matrix, ConfusionMatrixDisplay,
)
import matplotlib.pyplot as plt

import torch
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
    TrainingArguments,
    Trainer,
    DataCollatorWithPadding,
)
from datasets import Dataset
import warnings
warnings.filterwarnings("ignore")

print("✅ Imports complete")
print(f"PyTorch version: {torch.__version__}")
print(f"GPU available: {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
```

```
✅ Imports complete
PyTorch version: 2.11.0+cu128
GPU available: True
GPU: Tesla T4
```

Section 1: Load Your Dataset

Upload your labeled CSV and define your label map.

Your CSV must have at least two columns: `text` (the post/comment) and `label` (your string label).

```
# ── TODO ──────────────────────────────────────────────────────────────
# Define YOUR label map below.
# Keys are the string labels in your CSV; values are integers starting at 0.
# Add or remove entries to match your actual labels (2–4 labels supported).
#
# The example below is ILLUSTRATIVE ONLY (the r/nba taxonomy from the project
# page). DELETE it and use your own community's labels — submitting the
# example unchanged will not pass.
# ──────────────────────────────────────────────────────────────────────

LABEL_MAP = {
    "analysis":  0,   # ← Replace with your first label
    "hot_take":  1,   # ← Replace with your second label
    "reaction":  2,   # ← Replace with your third label (remove if you have 2 labels)
    # "fourth_label": 3,  # ← Uncomment if you have a fourth label
}

# ── END TODO ──────────────────────────────────────────────────────────

ID_TO_LABEL = {v: k for k, v in LABEL_MAP.items()}
NUM_LABELS = len(LABEL_MAP)
print(f"Labels: {LABEL_MAP}")
print(f"Number of labels: {NUM_LABELS}")
```

```
Labels: {'analysis': 0, 'hot_take': 1, 'reaction': 2}
Number of labels: 3
```

```
# Upload your CSV from your computer
from google.colab import files
print("Select your labeled dataset CSV file...")
uploaded = files.upload()
CSV_PATH = list(uploaded.keys())[0]
print(f"Uploaded: {CSV_PATH}")
```

```
Select your labeled dataset CSV file...
Choose Files takemeter.csv
takemeter.csv(text/csv) - 138126 bytes, last modified: 6/23/2026 - 100% done
Saving takemeter.csv to takemeter.csv
Uploaded: takemeter.csv
```

```
# Load and validate your dataset
df = pd.read_csv(CSV_PATH)

# ── TODO (if needed) ──────────────────────────────────────────────────
# If your CSV uses different column names, rename them here.
# Example: df = df.rename(columns={"post": "text", "category": "label"})
# ── END TODO ──────────────────────────────────────────────────────────

print(f"Columns: {df.columns.tolist()}")
print(f"Total examples: {len(df)}")
print()
print("Label distribution:")
print(df["label"].value_counts())

# Validate all labels are in LABEL_MAP
unknown = set(df["label"].unique()) - set(LABEL_MAP.keys())
if unknown:
    print(f"\n⚠️  Labels in CSV not found in LABEL_MAP: {unknown}")
    print("Update your LABEL_MAP above to include all labels.")
else:
    print("\n✅ All labels match your LABEL_MAP")

# Convert string labels to integers
df["label_id"] = df["label"].map(LABEL_MAP)
df = df.dropna(subset=["label_id"])
df["label_id"] = df["label_id"].astype(int)
```

```
Columns: ['text', 'label', 'notes', 'source', 'permalink', 'prelabeled']
Total examples: 219

Label distribution:
label
```

```
analysis    80
reaction    70
hot_take    69
Name: count, dtype: int64

✅ All labels match your LABEL_MAP
```

Section 2: Prepare Data for Training

Splits your dataset into train / validation / test sets and tokenizes the text.

```
# Train / val / test split — 70% / 15% / 15%
# Stratified so each split has roughly the same label distribution.
train_df, temp_df = train_test_split(
    df, test_size=0.30, random_state=42, stratify=df["label_id"]
)
val_df, test_df = train_test_split(
    temp_df, test_size=0.50, random_state=42, stratify=temp_df["label_id"]
)

print(f"Train: {len(train_df)} examples")
print(f"Validation: {len(val_df)} examples")
print(f"Test: {len(test_df)} examples")
print()
print("Train label distribution:")
print(train_df["label"].value_counts())
print()
print("Test label distribution:")
print(test_df["label"].value_counts())

# Reset indices (needed for clean HuggingFace Dataset conversion)
train_df = train_df.reset_index(drop=True)
val_df   = val_df.reset_index(drop=True)
test_df  = test_df.reset_index(drop=True)
```

```
Train: 153 examples
Validation: 33 examples
Test: 33 examples

Train label distribution:
label
analysis    56
reaction    49
hot_take    48
Name: count, dtype: int64

Test label distribution:
label
analysis    12
hot_take    11
reaction    10
Name: count, dtype: int64
```

```
# Load tokenizer and tokenize all splits
MODEL_NAME = "distilbert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)

def tokenize(examples):
    return tokenizer(examples["text"], truncation=True, max_length=256)

def make_dataset(df_split):
    ds = Dataset.from_pandas(
        df_split[["text", "label_id"]].rename(columns={"label_id": "labels"})
    )
    return ds.map(tokenize, batched=True)

train_dataset = make_dataset(train_df)
val_dataset   = make_dataset(val_df)
test_dataset  = make_dataset(test_df)

data_collator = DataCollatorWithPadding(tokenizer=tokenizer)
print("✅ Tokenization complete")
print(f"Sample keys: {list(train_dataset[0].keys())}")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_
```

config.json: 100% 483/483 [00:00<00:00, 38.1kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 4.39kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 757kB/s]

tokenizer.json: 100% 466k/466k [00:00<00:00, 1.93MB/s]

Map: 100% 153/153 [00:00<00:00, 1545.43 examples/s]

Map: 100% 33/33 [00:00<00:00, 807.67 examples/s]

Map: 100% 33/33 [00:00<00:00, 836.88 examples/s]

```
✅ Tokenization complete
Sample keys: ['text', 'labels', 'input_ids', 'token_type_ids', 'attention_mask']
```

Section 3: Fine-Tune Your Model

Loads `distilbert-base-uncased` with a classification head and fine-tunes it on your training data.
Training runs for 3 epochs and takes **5–15 minutes** on a T4 GPU.

> **Hyperparameter note:** The defaults below work well for datasets of 100–500 examples.
> If you change any values, note what you changed and why in your README.

```
# Load DistilBERT with a classification head
model = AutoModelForSequenceClassification.from_pretrained(
    MODEL_NAME,
    num_labels=NUM_LABELS,
    id2label=ID_TO_LABEL,
    label2id=LABEL_MAP,
)
print(f"✅ Model loaded: {MODEL_NAME}")
print(f"Output labels: {NUM_LABELS}")
```

model.safetensors: 100% 268M/268M [00:03<00:00, 824MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 3008.29it/s]

```
[transformers] DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_transform.bias    | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
classifier.bias         | MISSING    |
pre_classifier.weight   | MISSING    |
classifier.weight       | MISSING    |
pre_classifier.bias     | MISSING    |

Notes:
- UNEXPECTED:  can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstr
✅ Model loaded: distilbert-base-uncased
Output labels: 3
```

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return {"accuracy": accuracy_score(labels, predictions)}
```

```
# ── Hyperparameters ──────────────────────────────────────────────────
# num_train_epochs  — passes through the training data; 3 is a good default
#                     for small datasets. Increase cautiously; more epochs
#                     risk overfitting on 200 examples.
# learning_rate     — 2e-5 is the standard starting point for fine-tuning
#                     BERT-family models. Lower → slower but more stable.
# per_device_train_batch_size — 16 fits T4 GPU comfortably.
#                     Reduce to 8 if you get out-of-memory errors.
# ─────────────────────────────────────────────────────────────────────
```

```
training_args = TrainingArguments(
    output_dir="./takemeter-model",
    num_train_epochs=30,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=32,
    learning_rate=2e-5,
    weight_decay=0.01,
    warmup_steps=50,
    eval_strategy="epoch",
    save_strategy="epoch",
    save_total_limit=1,
    load_best_model_at_end=True,
    metric_for_best_model="accuracy",
    logging_steps=10,
    report_to="none",
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
)

print("Starting fine-tuning... (5–15 minutes on T4 GPU)")
trainer.train()
print("\n✅ Fine-tuning complete")
```

```
Starting fine-tuning... (5–15 minutes on T4 GPU)
```

[300/300 06:03, Epoch 30/30]

Epoch	Training Loss	Validation Loss	Accuracy
1	1.405796	1.459728	0.363636
2	1.408461	1.452386	0.363636
3	1.453229	1.440736	0.363636
4	1.459019	1.425023	0.363636
5	1.355875	1.405102	0.363636
6	1.411134	1.383222	0.363636
7	1.297214	1.361331	0.363636
8	1.320387	1.327867	0.363636
9	1.261778	1.285020	0.363636
10	1.244680	1.256540	0.363636
11	1.277178	1.244734	0.363636
12	1.167811	1.232589	0.363636
13	1.260127	1.224323	0.363636
14	1.229802	1.220041	0.363636
15	1.227011	1.218893	0.363636
16	1.235627	1.217178	0.363636
17	1.241134	1.216840	0.363636
18	1.243999	1.213893	0.363636
19	1.185353	1.212149	0.363636
20	1.210079	1.209203	0.363636

Section 4: Evaluate Fine-Tuned Model on Test Set

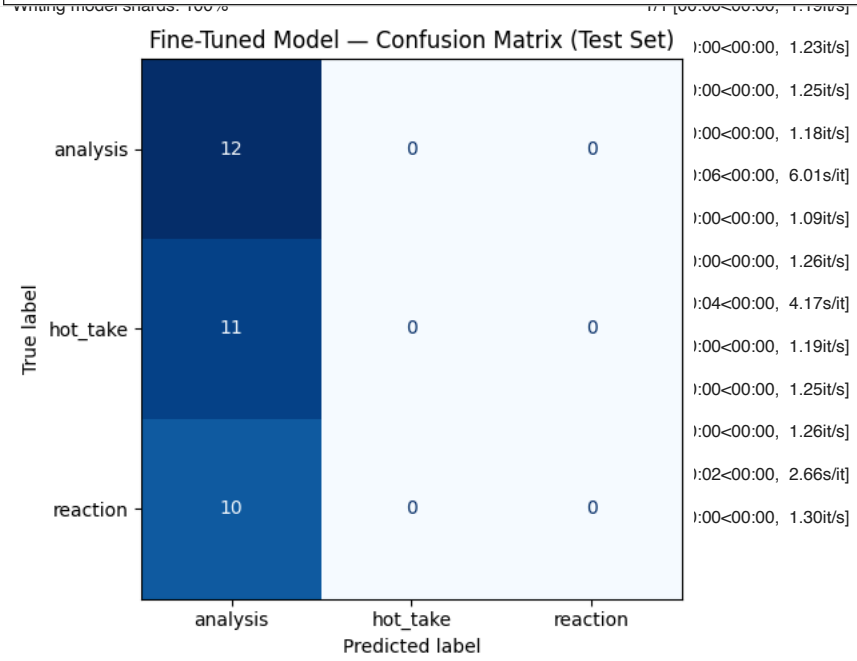
Runs inference on your locked test set and generates metrics and a confusion matrix.
These numbers go directly into your evaluation report.

```
# Run inference on the test set
print("Running inference on test set...")
ft_output = trainer.predict(test_dataset)
ft_pred_ids = np.argmax(ft_output.predictions, axis=-1)
ft_true_ids = ft_output.label_ids

ft_probs = torch.nn.functional.softmax(
    torch.tensor(ft_output.predictions), dim=-1
).numpy()

# Overall accuracy
ft_accuracy = accuracy_score(ft_true_ids, ft_pred_ids)
print(f"\n🎯 Fine-tuned model accuracy: {ft_accuracy:.3f}")

# Per-class metrics
label_names = [ID_TO_LABEL[i] for i in range(NUM_LABELS)]
print("\nPer-class metrics (fine-tuned model):")
print(classification_report(ft_true_ids, ft_pred_ids, target_names=label_names, zero_division=0))
```

```
Running inference on test set...
Writing model shards: 100%                                                      1/1 [00:07<00:00,  7.36s/it]
🎯 Fine-tuned model accuracy: 0.364
Writing model shards: 100%                                                      1/1 [00:00<00:00,  1.30it/s]

Per-class metrics (fine-tuned model):
Writing model shards: 100%                                                      1/1 [00:00<00:00,  1.11it/s]
              precision    recall  f1-score   support
Writing model shards: 100%                                                      1/1 [00:00<00:00,  1.30it/s]
    analysis       0.36      1.00      0.53        12
Writing model shards: 100%                                                      1/1 [00:05<00:00,  5.51s/it]
  comparison       0.00      0.00      0.00        11
    reaction       0.00      0.00      0.00        10
Writing model shards: 100%                                                      1/1 [00:05<00:00,  5.42s/it]
    accuracy                           0.36        33
Writing model shards: 100%                                                      1/1 [00:01<00:00,  1.00it/s]
   macro avg       0.12      0.33      0.18        33
weighted avg       0.13      0.36      0.19        33
Writing model shards: 100%                                                      1/1 [00:06<00:00,  6.21s/it]
```

```
# Confusion matrix
cm = confusion_matrix(ft_true_ids, ft_pred_ids)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=label_names)
fig, ax = plt.subplots(figsize=(7, 5))
disp.plot(ax=ax, cmap="Blues", colorbar=False)
```

```
ax.set_title("Fine-Tuned Model — Confusion Matrix (Test Set)")
plt.tight_layout()
plt.savefig("confusion_matrix.png", dpi=150)
plt.show()
print("✅ Saved: confusion_matrix.png  →  commit this to your repo and include in README")
```




```
):00<00:00,  1.23it/s]
):00<00:00,  1.25it/s]
):00<00:00,  1.18it/s]
):06<00:00,  6.01s/it]
):00<00:00,  1.09it/s]
):00<00:00,  1.26it/s]
):04<00:00,  4.17s/it]
):00<00:00,  1.19it/s]
):00<00:00,  1.25it/s]
):00<00:00,  1.26it/s]
):02<00:00,  2.66s/it]
):00<00:00,  1.30it/s]
```

✅ Saved: confusion_matrix.png → commit this to your repo and include in README

```
# Print wrong predictions for your error analysis
# Review these carefully — pick 3 to analyze in depth in your README.

wrong_idx = np.where(ft_pred_ids != ft_true_ids)[0]
print(f"Wrong predictions: {len(wrong_idx)} / {len(ft_true_ids)}\n")

for i, idx in enumerate(wrong_idx[:15]):
    text = test_df.iloc[idx]["text"]
    true_label = ID_TO_LABEL[ft_true_ids[idx]]
    pred_label = ID_TO_LABEL[ft_pred_ids[idx]]
    confidence = ft_probs[idx][ft_pred_ids[idx]]
    print(f"--- #{i+1} ---")
    print(f"Text:      {text[:200]}{'...' if len(text) > 200 else ''}")
    print(f"True:      {true_label}")
    print(f"Predicted: {pred_label}  (confidence: {confidence:.2f})")
    print()
```

```
Wrong predictions: 21 / 33

--- #1 ---
Text:      Rock snobs are the worst. Most of them over the years appear to have been unaware of how the 1950s "estal
True:      hot_take
Predicted: analysis  (confidence: 0.73)

--- #2 ---
Text:      You’re just not that interested in music which is okay. Most people aren’t really interested in music wh
True:      hot_take
Predicted: analysis  (confidence: 0.73)

--- #3 ---
Text:      Mainstream concerts have become so boring; it's practically just a choreographed routine, at least in po

      But also, add to that the fact that many people go...
True:      hot_take
Predicted: analysis  (confidence: 0.73)

--- #4 ---
Text:      I am not sure if this is the right sub to post this but I'll find someone to agree with me.

      Last night I went to a concert with 5 really popular artists (trappers and some reggaeton) in my ...
True:      hot_take
Predicted: analysis  (confidence: 0.73)

--- #5 ---
Text:      I just don't know that I'm in a place mentally to accept that someone unironically typed "so much gong" a
```

```
True:      hot_take
Predicted: analysis  (confidence: 0.73)

--- #6 ---
Text:      It is certainly one of them. But I am also a musician (ok…drummer) so the music itself was the memory. Bo
True:      reaction
Predicted: analysis  (confidence: 0.73)

--- #7 ---
Text:      I've always had the mindset that if a person truly claims to hate any one genre entirely then it's not re
True:      hot_take
Predicted: analysis  (confidence: 0.73)

--- #8 ---
Text:      Thta mixed with the vocal "once you free your mind of music being correct..." ahh! I'll never forget the
True:      reaction
Predicted: analysis  (confidence: 0.73)

--- #9 ---
Text:      The thunderbolt I experienced the first time I listened to The Wall was unbelievable.  I was 11, and it w
True:      reaction
Predicted: analysis  (confidence: 0.73)

--- #10 ---
Text:      You are allowed to voice your opinion. There is no obligation to preface every single thing you say with

      It sounds more like...
True:      hot_take
Predicted: analysis  (confidence: 0.73)
```

Section 5: Baseline Classifier (Groq)

Runs your zero-shot baseline using `llama-3.3-70b-versatile`.
You need to write the classification prompt using your label definitions.

```
from groq import Groq

# ── TODO: Add your Groq API key ──────────────────────────────────────────
# Recommended: use Colab Secrets so your key is never visible in the notebook.
#   1. Click the 🔑 icon in the left sidebar ("Secrets")
#   2. Add a secret named GROQ_API_KEY with your key as the value
#   3. Enable notebook access for the secret
#
# Then uncomment Option A below (and delete Option B).
#
# Option A — Colab Secrets (recommended):
from google.colab import userdata
GROQ_API_KEY = userdata.get("GROQ_API_KEY")
#
# Option B — paste directly (do not commit to GitHub):
#GROQ_API_KEY = "your_groq_api_key_here"

assert GROQ_API_KEY, (
    "GROQ_API_KEY not set — add it in the Colab Secrets panel (\U0001f511, left "
    "sidebar) and enable notebook access for this notebook, or use Option B above."
)

client = Groq(api_key=GROQ_API_KEY)
print("✅ Groq client initialized")
```

```
✅ Groq client initialized
```

```
# ── TODO: Write your classification prompt ─────────────────────────────
# Your prompt should:
#   1. Name your community and task
#   2. Define each label in plain language (copy from your planning.md)
#   3. Give one example post per label
#   4. Tell the model to output ONLY the label name — nothing else
#
# The model's response must match one of your label strings exactly,
# or the classify_with_groq() function below will mark it as unparseable.
#
# ──────────────────────────────────────────────────────────────────────
# REPLACE the placeholders below with your actual prompt. As written, this
# skeleton will NOT classify correctly — you must fill it in.
```

```
# SYSTEM_PROMPT = """
# You are classifying <posts> from <YOUR COMMUNITY>.
# Assign each post to exactly one of the following categories.

# <label_1>: <one-sentence definition of label 1>
# Example: "<one example post for label_1>"

# <label_2>: <one-sentence definition of label 2>
# Example: "<one example post for label_2>"

# <label_3>: <one-sentence definition of label 3>
# Example: "<one example post for label_3>"

# Respond with ONLY the label name.
# Do not explain your reasoning.

# Valid labels:
# <label_1>
# <label_2>
# <label_3>
# """


SYSTEM_PROMPT = """
  You are classifying comments and posts from r/LetsTalkMusic, a subreddit for in-depth music discussion. Assign
  each one to exactly one category based on HOW it supports its point.

  analysis: a structured argument about the music backed by specific, verifiable evidence
  (production/harmony/rhythm detail, historical or contextual fact, or named song/artist examples); the
  reasoning would stand even if the opinion framing were removed.
  Example: "Shoegaze isn't just a 'wall of noise' — My Bloody Valentine's Loveless uses the glide-guitar
  technique, detuning mid-chord with the tremolo arm, which is what creates that seasick pitch-bend. It's
  harmonically deliberate, not just distortion."

  hot_take: a bold, confident evaluative claim asserted without genuine support; it may sound credible but does
  not actually reason.
  Example: "Radiohead is the most overrated band of all time; people only like OK Computer to seem smart."

  reaction: an immediate emotional or personal response — a feeling or a memory — that makes no claim about the
  music's quality.
  Example: "First time hearing Carrie & Lowell and I'm literally sobbing, I don't have words."

  Decision rules for hard cases:
  - If a fact is cited but it is vague, cherry-picked, or decorative (there to sound credible while delivering a
  put-down), choose hot_take. Choose analysis only if it genuinely reasons.
  - An unsupported judgment about quality (e.g. "worst album ever") is hot_take. A pure feeling or personal
  experience with no quality claim is reaction.
  - If a musical detail only serves a feeling, choose reaction; if a feeling decorates a real argument, choose
  analysis.
  - When in doubt, prefer the lower-evidence label: reaction < hot_take < analysis.

  Respond with ONLY the label name, exactly as written here, using the underscore in hot_take: analysis,
  hot_take, or reaction. Do not explain.

  Valid labels:
  analysis
  hot_take
  reaction
  """

print("Prompt length:", len(SYSTEM_PROMPT), "characters")
```

```
Prompt length: 1961 characters
```

```
def classify_with_groq(text):
    """Classify a single post. Returns a label string or None if unparseable."""
    try:
        response = client.chat.completions.create(
            model="llama-3.3-70b-versatile",
            messages=[
                {"role": "system", "content": SYSTEM_PROMPT},
                {"role": "user", "content": f"Classify this post:\n\n{text}"},
            ],
            temperature=0,
            max_tokens=20,
        )
```

```
        raw = response.choices[0].message.content.strip().lower()
        # Match the model's output to a label. Check longest labels first so a
        # label that is a substring of another (e.g. "recommendation" vs.
        # "strong_recommendation") can't be matched by mistake.
        for label in sorted(LABEL_MAP, key=len, reverse=True):
            if raw == label or label in raw:
                return label
        return None  # model output didn't match any known label
    except Exception as e:
        print(f"API error: {e}")
        return None


# Run baseline on test set
print(f"Running baseline on {len(test_df)} examples...")
print("(May take a few minutes — 0.1s delay between requests to respect free-tier limits)\n")

baseline_preds = []
for i, (_, row) in enumerate(test_df.iterrows()):
    pred = classify_with_groq(row["text"])
    baseline_preds.append(pred)
    if (i + 1) % 10 == 0:
        print(f"  {i+1}/{len(test_df)} complete...")
    time.sleep(0.1)

none_count = baseline_preds.count(None)
if none_count > 0:
    print(f"\n⚠️  {none_count} responses could not be parsed.")
    print("Review your prompt — the model may not be outputting clean label names.")
```

```
Running baseline on 33 examples...
(May take a few minutes — 0.1s delay between requests to respect free-tier limits)

  10/33 complete...
  20/33 complete...
  30/33 complete...
```

```
# Baseline metrics (exclude unparseable responses)
valid = [(p, t) for p, t in zip(baseline_preds, test_df["label_id"])
         if p is not None]
bl_pred_ids = [LABEL_MAP[p] for p, _ in valid]
bl_true_ids = [t for _, t in valid]

bl_accuracy = accuracy_score(bl_true_ids, bl_pred_ids)
print(f"🎯 Baseline accuracy: {bl_accuracy:.3f}  "
      f"(evaluated on {len(valid)}/{len(test_df)} parseable responses)")
print()
label_names = [ID_TO_LABEL[i] for i in range(NUM_LABELS)]
print("Per-class metrics (baseline):")
print(classification_report(bl_true_ids, bl_pred_ids, target_names=label_names, zero_division=0))
```

```
🎯 Baseline accuracy: 0.909  (evaluated on 33/33 parseable responses)

Per-class metrics (baseline):
              precision    recall  f1-score   support

    analysis       1.00      0.92      0.96        12
    hot_take       0.90      0.82      0.86        11
    reaction       0.83      1.00      0.91        10

    accuracy                           0.91        33
   macro avg       0.91      0.91      0.91        33
weighted avg       0.92      0.91      0.91        33
```

Section 6: Compare Results and Export

Side-by-side comparison of both models.
Download the output files and commit them to your GitHub repo.

```
print("=" * 50)
print("RESULTS COMPARISON")
print("=" * 50)
print(f"{'Model':<35} {'Accuracy':>8}")
print("-" * 45)
```